

JEE-MAINS-2020-JAN-7-SHIFT-2 Final

PHYSICS

Question 1: In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is

Options:

- (a) 4.9 mm
- (b) 3.9 mm
- (c) 5.9 mm
- (d) 6.9 mm

Question 2: A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears 2 beats/sec. The oscillation frequency of each tuning fork is 0 = 1400 Hz and the velocity of sound in air is 350 m/s. The speed of each tuning fork is close to

Options:

- (a) $\frac{1}{4}$ m/s
- (b) $\frac{1}{2}$ m/s
- (c) 1 m/s
- (d) $\frac{1}{8}$ m/s

Question 3: An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator ($g = 10 \text{ m/s}^2$) must be at least

Options:

- (a) 62360 W
- (b) 56300 W
- (c) 48000 W
- (d) 66000 W

Question 4: Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperatures, T_1 and T_2 . The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2 . T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2 , if both the engines perform equal amount of work ?

Options:

- (a) $T = \sqrt{T_1 T_2}$
- (b) $T = \frac{T_1 + T_2}{2}$

(c) $T = \frac{2T_1T_2}{T_1 + T_2}$

(d) $T = 0$

Question 5: An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is

Options:

(a) $\frac{9}{16}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{81}{256}$

(d) $\frac{3}{4}$

Question 6: Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecules changes from τ_1 to τ_2 . If $C_P/C_V = \gamma$ for this gas then a good estimate for τ_1/τ_2 is given by

Options:

(a) $\left(\frac{1}{2}\right)^\gamma$

(b) 2

(c) $\frac{1}{2}$

(d) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$

Question 7: A planar loop of wire rotates in a uniform magnetic field. Initially at $t = 0$, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10 s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at

Options:

(a) 5.0 s and 7.5 s

(b) 5.0 s and 10.0 s

(c) 2.5 s and 7.5 s

(d) 2.5 s and 5.0 s

Question 8: The electric field of a plane electromagnetic wave is given by

$\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$. At $t = 0$, a positively charged particle is at the point

$(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at $(t = 0)$ is $v_0 \hat{k}$, the

force acting on it due to the wave is

Options:

- (a) Zero
- (b) Parallel to $\hat{k}$
- (c) antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$
- (d) Parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Question 9: In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of building will be

Options:

- (a) 20 A
- (b) 15 A
- (c) 10 A
- (d) 25 A

Question 10: A thin lens made of glass (refractive index = 1.5) of focal length $f = 16$ cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_l , then the ratio f_l/f is closest to the integer

Options:

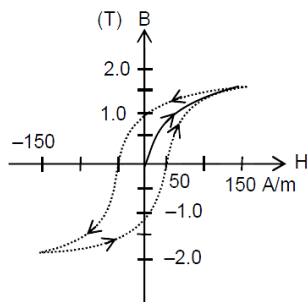
- (a) 1
- (b) 5
- (c) 17
- (d) 9

Question 11: An emf of 20 V is applied at time $t = 0$ to a circuit containing in series 10 mH inductor and 5Ω resistor. The ratio of the currents at time $t = \infty$ and at $t = 40$ s is close to : (Take $e^2 = 7.389$)

Options:

- (a) 1.46
- (b) 1.06
- (c) 0.84
- (d) 1.15

Question 12: The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are



Options:

- (a) 1.0 T, 50 A/m and 1.5 T
- (b) 150 A/m, 1.0 T and 1.5 T
- (c) 1.5 T, 50 A/m and 1.0 T
- (d) 1.5T, 50 A/m and 1.0 T

Question 13: A mass of 10 kg is suspended by a rope of length 4 m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals : (Take $g = 10 \text{ ms}^{-2}$ and the rope to be massless)

Options:

- (a) 75 N
- (b) 90 N
- (c) 100 N
- (d) 70 N

Question 14: The dimensions of $\frac{B^2}{2\mu_0}$ where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is

Options:

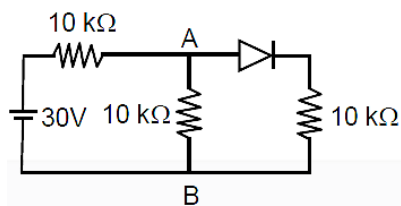
- (a) MLT^{-2}
- (b) $\text{ML}^{-1}\text{T}^{-2}$
- (c) ML^2T^{-1}
- (d) ML^2T^{-2}

Question 15: A box weighs 196 N on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take $g = 10 \text{ ms}^{-2}$ at the north pole and the radius of the earth = 6400 km)

Options:

- (a) 194.66 N
- (b) 195.66 N
- (c) 195.32
- (d) 194.32 N

Question 16: In the figure, potential difference between A and B is



Options:

- (a) 15 V
- (b) Zero
- (c) 10 V
- (d) 5 V

Question 17: A particle of mass m and charge q has an initial velocity $\vec{v} = v_o \hat{j}$. If an electric field $\vec{E} = E_o \hat{i}$ and magnetic field $\vec{B} = B_o \hat{i}$ act on the particle, its speed will double after a time

Options:

- (a) $\frac{2mv_o}{qE_o}$
- (b) $\frac{3mv_o}{qE_o}$
- (c) $\frac{\sqrt{2}mv_o}{qE_o}$
- (d) $\frac{\sqrt{3}mv_o}{qE_o}$

Question 18: Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is

Options:

- (a) $2\pi a^4 \left(\frac{aA}{4} + \frac{B}{5} \right)$
- (b) $2\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$
- (c) $\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$
- (d) $2\pi a^4 \left(\frac{A}{4} + \frac{B}{5} \right)$

Question 19: An electron (of mass m) and a photon have the same energy E in the range of a few eV. The ratio of the de-Broglie wavelength associated with the electron and the wavelength of the photon is (c = speed of light in vacuum)

Options:

(a) $\frac{1}{c} \left(\frac{2E}{m} \right)^{\frac{1}{2}}$

(b) $\frac{1}{c} \left(\frac{E}{2m} \right)^{\frac{1}{2}}$

(c) $c(2mE)^{\frac{1}{2}}$

(d) $\frac{1}{c} \left(\frac{E}{2m} \right)^{\frac{1}{2}}$

Question 20: The activity of a radioactive sample falls from 700 s^{-1} to 500 s^{-1} in 30 minutes. Its half life is close to

Options:

- (a) 62 min
- (b) 66 min
- (c) 72 min
- (d) 52 min

Question 21: The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $N/10 \Omega$, where N is an integer then value of N is _____.

Question 22: M grams of steam at 100°C is mixed with 200g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is_____.

Question 23: Consider uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value of $100 \times b/a$ for box not to topple before moving is

Question 24: A 60 pF capacitor is fully charged by a 20V supply. It is then disconnected from the supply and is connected to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ)_____.

Question 25: The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$ The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is

CHEMISTRY

Question 26: Which of the following statements is correct?

Options:

- (a) Gluconic acid is a dicarboxylic acid
- (b) Gluconic acid is a partial oxidation product of glucose
- (c) Gluconic acid can form cyclic (acetal/hemiacetal) structure
- (d) Gluconic acid is obtained by oxidation of glucose with HNO_3

Question 27: Two open beakers one containing a solvent and the other containing a mixture of that solvent with a non-volatile solute are together sealed in a container. Over time

Options:

- (a) the volume of the solution decreases and the volume of the solvent increases
- (b) the volume of the solution and the solvent does not change
- (c) the volume of the solution increases and the volume of the solvent decreases
- (d) the volume of the solution does not change and the volume of the solvent decreases

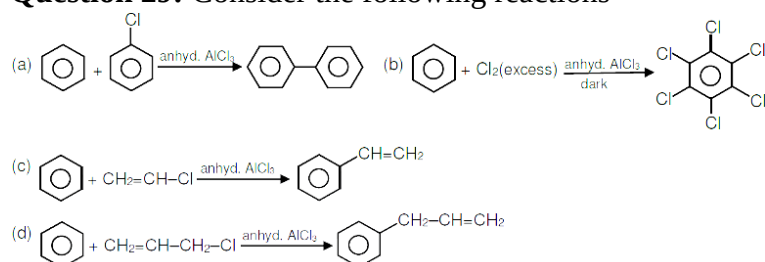
Question 28: For the reaction, $2\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$; the observed rate expression is, $\text{rate} = k_f[\text{NO}]^2 [\text{H}_2]$.

The rate expression for the reverse reaction is

Options:

- (a) $k_b[\text{N}_2][\text{H}_2\text{O}]$
- (b) $k_b[\text{N}_2][\text{H}_2\text{O}]^2/[\text{NO}]$
- (c) $k_b[\text{N}_2][\text{H}_2\text{O}]^2/[\text{H}_2]$
- (d) $k_b[\text{N}_2][\text{H}_2\text{O}]^2$

Question 29: Consider the following reactions



Which of these reactions are possible?

Options:

- (a) (a) and (d)
- (b) (a) and (b)
- (c) (b) and (d)
- (d) (b), (c) and (d)

Question 30: The bond order and the magnetic characteristics of CN^- are

Options:

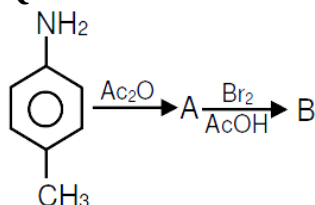
- (a) $2\frac{1}{2}$, diamagnetic
- (b) 3, paramagnetic
- (c) 3, diamagnetic
- (d) $2\frac{1}{2}$, paramagnetic

Question 31: The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is

Options:

- (a) 0 and 0
- (b) 2 and 2
- (c) 0 and 2
- (d) 0 and 1

Question 32: In the following reaction sequence,



the major product B is :

Options:

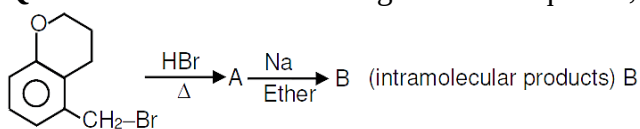
- (a)
- (b)
- (c)
- (d)

Question 33: In the following reactions, products (A) and (B), respectively, are
 $\text{NaOH} + \text{Cl}_2$ (A) + side products (hot and conc.)
 $\text{Ca(OH)}_2 + \text{Cl}_2$ (B) + side products (dry)

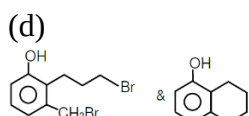
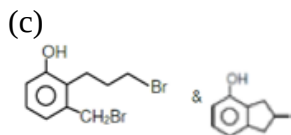
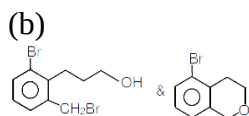
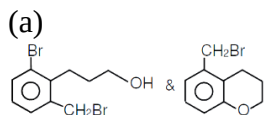
Options:

- (a) NaClO_3 and $\text{Ca(ClO}_3)_2$
- (b) NaOCl and Ca(OCl)_2
- (c) NaOCl and $\text{Ca(ClO}_3)_2$
- (d) NaClO_3 and Ca(OCl)_2

Question 34: In the following reaction sequence, structures of A and B, respectively will be



Options:

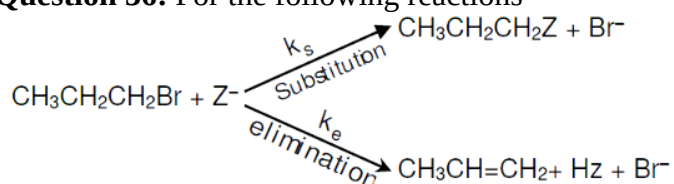


Question 35: The refining method used when the metal and the impurities have low and high melting temperatures, respectively, is

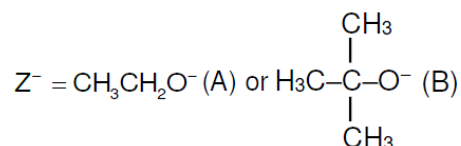
Options:

- (a) zone refining
- (b) vapour phase refining
- (c) liquation
- (d) distillation

Question 36: For the following reactions



Where

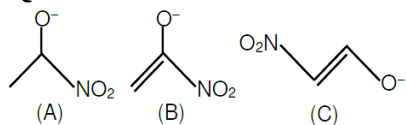


k_s and k_e , are, respectively, the rate constants for substitution and elimination, and $\mu = K_s/K_e$, the correct option is _____.

Options:

- (a) $\mu_A > \mu_B$ and $k_e(\text{B}) > k_e(\text{A})$
- (b) $\mu_A > \mu_B$ and $k_e(\text{A}) > k_e(\text{B})$
- (c) $\mu_B > \mu_A$ and $k_e(\text{B}) > k_e(\text{A})$
- (d) $\mu_B > \mu_A$ and $k_e(\text{A}) > k_e(\text{B})$

Question 37: The correct order of stability for the following alkoxides is



Options:

- (a) (C) > (B) > (A)

- (b) $(B) > (C) > (A)$
- (c) $(C) > (A) > (B)$
- (d) $(B) > (A) > (C)$

Question 38: Within each pair of element F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are

Options:

- (a) Cl, S and Li
- (b) F, S and Li
- (c) F, Se and Na
- (d) Cl, Se and Na

Question 39: The ammonia (NH_3) released on quantitative reaction of 0.6 g urea (NH_2CONH_2) with sodium hydroxide (NaOH) can be neutralized by

Options:

- (a) 200 ml of 0.4 N HCl
- (b) 200 ml of 0.2 N HCl
- (c) 100 ml of 0.1 N HCl
- (d) 100 ml of 0.2 N HCl

Question 40: A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane: ethyl acetate (20 : 80), the sequence of obtained compounds is

Options:

- (a) (B), (C) and (A)
- (b) (A), (B) and (C)
- (c) (B), (A) and (C)
- (d) (C), (A) and (B)

Question 41: The equation that is incorrect is

Options:

- (a) $(A_m^\circ)_{\text{NaBr}} - (A_m^\circ)_{\text{NaI}} = (A_m^\circ)_{\text{KBr}} - (A_m^\circ)_{\text{NaBr}}$
- (b) $(A_m^\circ)_{\text{H}_2\text{O}} = (A_m^\circ)_{\text{HCl}} + (A_m^\circ)_{\text{NaOH}} - (A_m^\circ)_{\text{NaCl}}$
- (c) $(A_m^\circ)_{\text{KCl}} - (A_m^\circ)_{\text{NaCl}} = (A_m^\circ)_{\text{KBr}} - (A_m^\circ)_{\text{NaBr}}$
- (d) $(A_m^\circ)_{\text{NaBr}} - (A_m^\circ)_{\text{NaCl}} = (A_m^\circ)_{\text{KBr}} - (A_m^\circ)_{\text{KCl}}$

Question 42: Among the statements(a)-(d), the incorrect ones are :

- (a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments
- (b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{eg}^4 e_g^2$
- (c) Wavelength of light absorbed by $[\text{Co}(\text{en})_3]^{3+}$ is lower than that of $[\text{CoF}_6]^{3-}$
- (d) If the Δ_0 for an octahedral complex of Co(III) is $18,000 \text{ cm}^{-1}$, the Δ_t for its tetrahedral complex with the same ligand will be $16,000 \text{ cm}^{-1}$

Options:

- (a) (b) and (c) only
- (b) (a) and (b) only
- (c) (c) and (d) only
- (d) (a) and (d) only

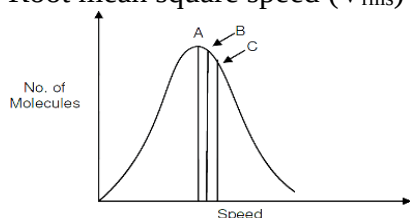
Question 43: The redox reaction among the following is

Options:

- (a) reaction of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ with AgNO_3
- (b) formation of ozone from atmospheric oxygen in the presence of sunlight
- (c) combination of dinitrogen with dioxygen at 2000 K
- (d) reaction of H_2SO_4 with NaOH

Question 44: Identify the correct labels of A, B and C in the following graph from the options given below :

Root mean square speed (V_{rms}) ; most probable speed (V_{mp}); Average speed (V_{av})



Options:

- (a) A – V_{mp} ; B – V_{rms} ; C – V_{av}
- (b) A – V_{rms} ; B – V_{mp} ; C – V_{av}
- (c) A – V_{av} ; B – V_{rms} ; C – V_{mp}
- (d) A – V_{mp} ; B – V_{av} ; C – V_{rms}

Question 45: Among statements (a)-(d), the correct ones are :

- (a) Decomposition of hydrogen peroxide gives dioxygen
- (b) Like hydrogen peroxide, compounds, such as KClO_3 , $\text{Pb}(\text{NO}_3)_2$ and NaNO_3 when heated liberate dioxygen.
- (c) 2-Ethylanthraquinone is useful for the industrial preparation of hydrogen peroxide.
- (d) Hydrogen peroxide is used for the manufacture of sodium perborate.

Options:

- (a) (a), (c) and (d) only
- (b) (a), (b), (c) and (d)
- (c) (a), (b) and (c) only
- (d) (a) and (c) only

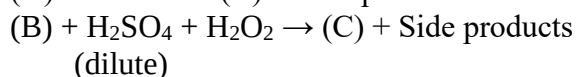
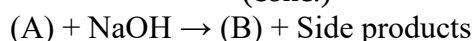
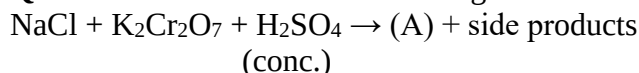
Question 46: The number of sp^2 hybridised carbons present in "Aspartame" is _____

Question 47: 3g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is _____

Question 48: The standard heat of formation ($\Delta_f H_{298}^\circ$) of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are -1560 , -393.5 and -286 kJ/mol, respectively is _____

Question 49: The flocculation value of HCl for arsenic sulphide sol. is 30 m mol L^{-1} . if H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams, of H_2SO_4 in 250 ml required for the above purpose is _____ (Molecular mass of $\text{H}_2\text{SO}_4 = 98 \text{ g/mol}$)

Question 50: Consider the following reactions:



The sum of the total number of atoms in one molecule each of (A), (B) and (C) is _____

MATHEMATICS

Question 51: The locus of the mid-point of the perpendiculars drawn from points on the line, $x = 2y$ to the line $x = y$ is

Options:

(a) $3x - 2y = 0$

(b) $3x - 3y = 0$

(c) $5x - 7y = 0$

(d) $7x - 5y = 0$

Question 52: The area (in sq. units) of the region $\{(x,y) \in \mathbb{R}^2 | 4x^2 \leq y \leq 8x + 12\}$ is

Options:

(a) $\frac{127}{3}$

(b) $\frac{125}{3}$

(c) $\frac{128}{3}$

(d) $\frac{124}{3}$

Question 53: If $\frac{3+i \sin \theta}{4-i \cos \theta}$, $\theta \in [0, 2\pi]$, is a real number, then an argument of

$\sin \theta + i \cos \theta$ is

Options:

(a) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$

(b) $\tan^{-1}\left(\frac{4}{3}\right)$

(c) $\pi - \tan^{-1}\left(\frac{4}{3}\right)$

(d) $\tan^{-1}\left(\frac{3}{4}\right)$

Question 54: In a workshop there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $(\frac{3}{4})^3 k$, then k is equal to

Options:

(a) $\frac{17}{2}$

(b) $\frac{17}{4}$

(c) $\frac{17}{8}$

(d) 4

Question 65: The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is

Options:

(a) $\frac{4 - \sqrt{5}}{3}$

(b) $\frac{\sqrt{7} - 2}{3}$

(c) $\frac{2}{3}$

(d) $\frac{4 - \sqrt{7}}{3}$

Question 56: Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true?

Options:

(a) $p_5 = p_2 \cdot p_3$

(b) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(c) $p_3 = p_5 - p_4$

(d) $p_5 = 11$

Question 57: Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the point A and B. The $(AB)^2$ is equal to

Options:

(a) $\frac{52}{5}$

(b) $\frac{32}{5}$

- (c) $\frac{56}{5}$
(d) $\frac{64}{5}$

Question 58: The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is

Options:

- (a) $\log_e \left(\frac{3}{2} \right)$
(b) $\log_e \left(\frac{4}{3} \right)$
(c) $\log_e \sqrt{2}$
(d) $\log_e 2$

Question 59: Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to:

Options:

- (a) $\frac{\sqrt{5}}{2}$
(b) $-\frac{\sqrt{5}}{2}$
(c) $\frac{2}{\sqrt{5}}$
(d) $-\frac{\sqrt{5}}{4}$

Question 60: The number of ordered pairs (r, k) for which $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1}$, where k is an integer, is

Options:

- (a) 3
(b) 6
(c) 4
(d) 2

Question 61: Let $a_1, a_2, a_3, \dots$ be a G.P. such that $a_1 < 0$, $a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If

$$\sum_{i=1}^9 a_i = 4\lambda \text{ then } \lambda =$$

Options:

- (a) 171
(b) -513

- (c) $\frac{511}{3}$
(d) -171

Question 62: Let A, B, C and D be four non-empty sets. The contrapositive statement of "If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$ " is

Options:

- (a) If $A \not\subseteq C$, then $A \subseteq B$ and $B \subseteq D$
(b) If $A \not\subseteq C$, then $A \not\subseteq B$ and $B \subseteq D$
(c) If $A \subseteq C$, then $B \subseteq A$ or $D \subseteq B$
(d) If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$

Question 63: If θ_1 and θ_2 be respectively the smallest and the largest values of θ in

$(0, 2\pi) - (\pi)$ which satisfy the equation, $2 \cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0$, then $\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is equal to

Options:

- (a) $\frac{\pi}{9}$
(b) $\frac{\pi}{3} + \frac{1}{6}$
(c) $\frac{2\pi}{3}$
(d) $\frac{\pi}{3}$

Question 64: Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If

$\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then the ordered pair, $(\lambda, \vec{d})$ is equal to

Options:

- (a) $\left(\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$
(b) $\left(-\frac{3}{2}, 3\vec{c} \times \vec{b}\right)$
(c) $\left(\frac{3}{2}, 3\vec{b} \times \vec{c}\right)$
(d) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$

Question 65: Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If

$\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3}\right) = 4$ then which one of the following is not true?

Options:

- (a) $x = 1$ is a point of minima and $x = -1$ is a point of maxima of f .
- (b) $f(1) - 4f(-1) = 4$.
- (c) $x = 1$ is a point of maxima and $x = -1$ is a point of minimum of f .
- (d) f is an odd function.

Question 66: if $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in \mathbb{R}$, then the distance between the foci of the ellipse is

Options:

- (a) $2\sqrt{2}$
- (b) $2\sqrt{7}$
- (c) 4
- (d) $2\sqrt{5}$

Question 67: Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x) \frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x -axis at a point whose abscissa is

Options:

- (a) $2 - e$
- (b) $2 + e$
- (c) $-e$
- (d) 2

Question 68: Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)} a_{ji}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then the determinant of A is

Options:

- (a) $\frac{1}{9}$
- (b) $\frac{1}{81}$
- (c) 3
- (d) $\frac{1}{3}$

Question 69: If the sum of the first 40 terms of the series, $3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \dots$ is $(102)m$, then m is equal to

Options:

- (a) 25
- (b) 20
- (c) 10
- (d) 5

Question 70: The coefficient of x^7 in the expression $(1 + x)^{10} + x(1 + x)^9 + x^2(1 + x)^8 + \dots + x^{10}$ is

Options:

- (a) 210
- (b) 330
- (c) 420
- (d) 120

Question 71: If the foot of the perpendicular drawn from the point $(1, 0, 3)$ on a line passing through $(\alpha, 7, 1)$ is $(5/3, 7/3, 17/3)$, then α is equal to

Question 72: If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+3x}{1-2x} \right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$

is continuous, then k is equal to

Question 73: If the mean and variance of eight numbers 3, 7, 9, 12, 13, 20, x and y be 10 and 25 respectively, then $x \cdot y$ is equal to _____.

Question 74: If the system of linear equations,

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$3x + 2y + \lambda z = \mu$$

has more than two solutions, then $\mu - \lambda^2$ is equal to _____.

Question 75: Let $X = \{n \in \mathbb{N} : 1 \leq n \leq 50\}$. If $A = \{n \in X : n \text{ is a multiple of } 2\}$; $B = \{n \in X : n \text{ is a multiple of } 7\}$, then the number of elements in the smallest subset of X containing both A and B is _____.

Answer Key:

1. (d)	2. (a)	3. (d)	4. (a)	5. (a)	6. (c)	7. (d)	8. (c)	9. (a)	10. (c)
11. (d)	12. (c)	13. (b)	14. (a)	15. (d)	16. (b)	17. (b)	18. (c)	19. (d)	20. (b)
21. (6.00)	22. (50.00)	23. (40.00)	24. (12.00)	25. (90.00)	26. (b)	27. (c)	28. (c)	29. (c)	30. (c)
31. (a)	32. (c)	33. (d)	34. (d)	35. (c)	36. (a)	37. (a)	38. (a)	39. (d)	40. (d)
41. (a)	42. (c)	43. (c)	44. (d)	45. (b)	46. 9	47. (5.22)	48. – (192.50)	49. (0.38)	50. (18.00)
51. (c)	52. (c)	53. (c)	54. (c)	55. (d)	56. (a)	57. (d)	58. (d)	59. (b)	60. (c)
61. (d)	62. (d)	63. (d)	64. (d)	65. (a)	66. (b)	67. (a)	68. (a)	69. (b)	70. (b)
71. (4.00)	72. (5.00)	73. (54.00)	74. (13.00)	75. (29.00)					

SOLUTIONS:

PHYSICS

Solution 1:

The separation between the successive bright fringes on the screen is

$$\beta = \frac{\lambda D}{d} = \frac{589 \times 10^{-9} \times 1.5}{0.15 \times 10^{-3}} = 5.9 \text{ mm}$$

Solution 2:

The speed of each tuning fork is close to



$$f_0 \left(\frac{C}{C-V} \right) - f_0 \left(\frac{C}{C+V} \right) = 2$$

$$V = \frac{1}{4} \text{ m/s}$$

Solution 3:

Net force on motor will be

$$F_m = [920 + 68(10)]g + 6000$$

$$= 22000 \text{ N}$$

$$P_m = \vec{F}_m \cdot \vec{v}$$

$$= 22000 \times 3$$

$$= 66000 \text{ watt}$$

Solution 4:

How is T related to T_1 and T_2 , if both the engines perform equal amount of work ?

$$Q_H - Q_L = Q_L - Q_L'$$

$$2Q_L = Q_H + Q_L'$$

$$2 = \frac{T_1}{T} + \frac{T_2}{T}$$

$$T = \frac{T_1 + T_2}{2}$$

Solution 5:

$$A_1 V_1 = A_2 V_2$$

$$\frac{V_1}{V_2} = \frac{A_2}{A_1} = \left(\frac{4.8}{6.4}\right)^2 = \frac{9}{16}$$

Solution 6:

If $C_p/C_v = \gamma$ for this gas then a good estimate for τ_1/τ_2 is given by

$$\text{Relaxation time } (\tau) \propto \frac{V}{\sqrt{T}}$$

$$\text{And } T \propto \frac{1}{V^{\gamma-1}}$$

$$\tau \propto V^{1+\frac{\gamma-1}{2}}$$

$$\tau \propto V^{\frac{1+\gamma}{2}}$$

$$\frac{\tau_f}{\tau_i} = \left(\frac{2V}{V}\right)^{\frac{1+\gamma}{2}}$$

$$\frac{\tau_f}{\tau_i} = (2)^{\frac{1+\gamma}{2}}$$

Solution 7:

If it rotates with a period of 10 s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at

$$\therefore \omega = \frac{2\pi}{T} = \frac{\pi}{5}$$

$$\text{When } \omega t = \frac{\pi}{2}$$

$\therefore \phi$ will be minimum

$\therefore e$ will be maximum

$$\text{When } \omega t = \pi$$

$\therefore \phi$ will be maximum

$\therefore e$ will be minimum

$$t = \frac{\pi}{\frac{\pi}{5}} = 5 \text{ sec}$$

Solution 8:

Force due to electric field is in direction $-\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$

because at $t = 0$, $E = -\frac{(\hat{i} + \hat{j})}{\sqrt{2}} E_0 \therefore$ Net force is antiparallel to $\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$

Solution 9:

Total power is $(15 \times 45) + (15 \times 100) + (15 \times 10) + (2 \times 1000) = 4325 W$

$$I = \frac{4325}{220} = 19.66 A$$

Answer is 20 Amp.

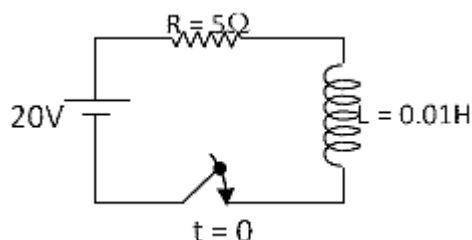
Solution 10:

If its focal length in liquid is f , then the ratio f_t/f is closest to the integer

$$\begin{aligned} \frac{1}{f_a} &= \left(\frac{\mu_g}{\mu_a} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ \frac{1}{f_m} &= \left(\frac{\mu_g}{\mu_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ \Rightarrow \frac{f_a}{f_m} &= \frac{\left(\frac{\mu_g}{\mu_m} - 1 \right)}{\left(\frac{\mu_g}{\mu_a} - 1 \right)} = \frac{\left(\frac{1.50}{1.42} - 1 \right)}{\left[\frac{1.50}{1} - 1 \right]} = \frac{0.08}{(1.92)(0.5)} \\ \Rightarrow \frac{f_m}{f_a} &= \frac{(1.42)(0.5)}{0.08} = 8.875 \approx 9 \end{aligned}$$

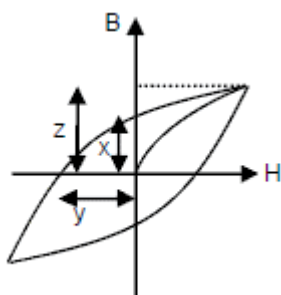
Solution 11:

$$i = i_0 \left(1 - e^{\frac{-t}{L/R}} \right)$$



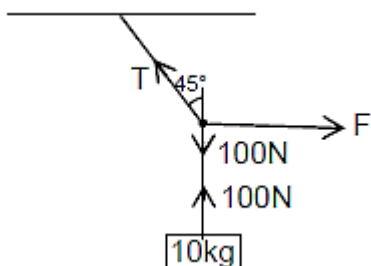
$$\begin{aligned} &= \frac{20}{5} \left(1 - e^{\frac{-t}{0.01/5}} \right) \\ &= 4 \left(1 - e^{-500t} \right) \\ i_{\infty} &= 4 \\ i_{40} &= 4 \left(1 - e^{-500 \times 40} \right) \\ &= 4 \left(1 - \frac{1}{(e^2)^{10000}} \right) = 4 \left(1 - \frac{1}{7.29^{10000}} \right) \\ \frac{i_{\infty}}{i_{40}} &\approx 1 \text{ greater than one.} \end{aligned}$$

Solution 12:



Solution 13:

F equals :



$$\frac{T}{\sqrt{2}} = 100$$

$$\frac{T}{\sqrt{2}} = F$$

$$F = 100 \text{ N}$$

Solution 14:

$$\text{Energy density} = \frac{B^2}{2\mu_0}$$

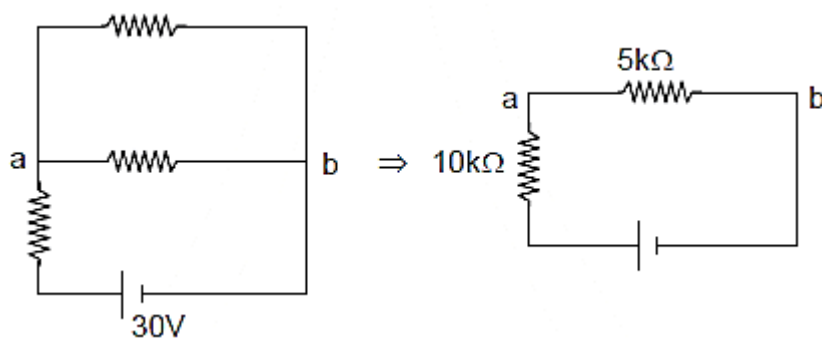
$$= \frac{\text{Force} \times \text{displacement}}{(\text{displacement})^3} = \frac{MLT^{-2} \cdot L}{L^3} = ML^{-1}T^{-2}$$

Solution 15:

$$\text{at equator, weight} = mg - m\omega^2 R$$

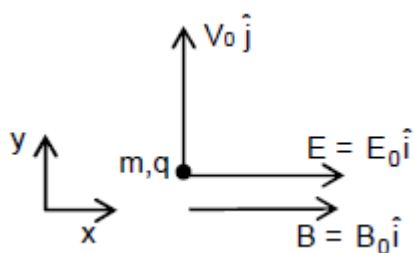
$$= 196 - (19.6) \left[\frac{2\pi}{24 \times 3600} \right]^2 \times 6400 \times 10^3 = 195.33 \text{ N}$$

Solution 16:



$$\text{So, } V_{ab} = \frac{30}{5+10} \times 5 = 10V$$

Solution 17:

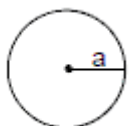


As $\vec{v} = v_0 \hat{j}$ (magnitude of velocity does not change in y-z plane)

$$\therefore \sqrt{3}v_0 = 0 + \frac{qE}{m}t; \quad t = \frac{mv_0\sqrt{3}}{qE}$$

Solution 18:

The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is



$$\sigma = A + Br$$

$$\int dm = \int (A + Br) 2\pi r dr$$

$$I = \int dm r^2$$

$$= \int_0^a (A + Br) 2\pi r^3 dr$$

$$= 2\pi \left(A \frac{a^4}{4} + B \frac{a^5}{5} \right) = 2\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$$

Solution 19:

$$\lambda_d \text{ for electron} = \frac{h}{\sqrt{2mE}}$$

$$\text{Ratio} = \frac{h}{\sqrt{2mE}} \frac{E}{hC} = \frac{1}{C} \sqrt{\frac{E}{2m}}$$

Solution 20:

$$\ln \left[\frac{A_0}{A_t} \right] = \lambda t$$

$$\Rightarrow \ln 2 = \lambda t_{1/2} \quad \dots(i)$$

$$\Rightarrow \ln \left(\frac{700}{500} \right) = \lambda (30 \text{ min}) \quad \dots(ii)$$

Dividing (i) and (ii)

$$\Rightarrow \frac{\ln 2}{\ln(7/5)} = \frac{t_{1/2}}{(30 \text{ min})} \Rightarrow (2.06004) 30 = t_{1/2} = 61.8 \text{ min.}$$

Solution 21:

$$\varepsilon = 560x \quad \dots(1)$$

$$\frac{\varepsilon \times 10}{10 + r} = 500x \quad \dots(2)$$

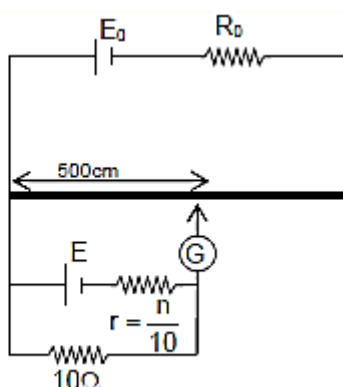
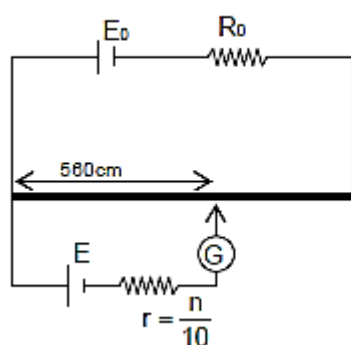
from (1) and (2)

$$\frac{560 \times 10}{10 + r} = 500s$$

$$56 = 50 + 5r$$

$$r = \frac{6}{5} = 1.2 \Omega$$

$$n = 12$$



Solution 22:

$$M_{\text{ice}} L_f + m_{\text{ice}} (40 - 0) C_w = m_{\text{steam}} L_v + m_{\text{steam}} (100 - 40) C_w$$

$$\Rightarrow 200 (120) = m(600)$$

$$m = 40 \text{ gm}$$

Solution 23:

For no toppling

$$F\left(\frac{a}{2} + b\right) \leq mg \frac{a}{2}$$

$$\mu \frac{a}{2} + \mu b \leq \frac{a}{2}$$

$$0.2a + 0.4b \leq 0.5a$$

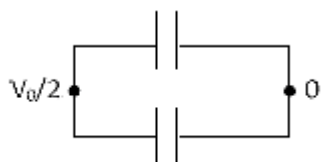
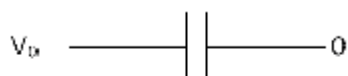
$$0.4b \leq 0.3a$$

$$b \leq 0.75a$$

But it is not possible as maximum value of b can be equal to 0.5a only

$$\therefore \left(100 \frac{b}{a}\right)_{\max.} = 50.00$$

Solution 24:



$$V_0 = 20V$$

Heat loss

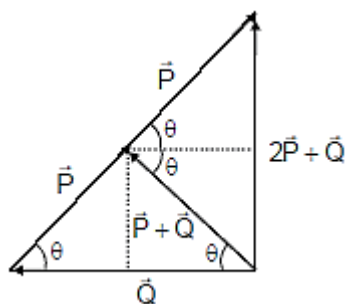
$$= \frac{1}{2} CV_0^2 - 2 \left[\frac{1}{2} C \left(\frac{V_0}{2} \right)^2 \right]$$

$$= \frac{CV_0^2}{4}$$

$$= \frac{(60 \times 10^{-12})(20)^2}{4} J$$

$$= 6 \times 10^{-9} J = 6 nJ$$

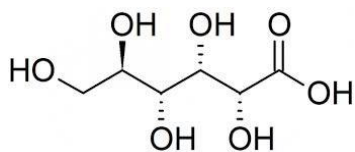
Solution 25:



Angle between $(2\vec{P} + \vec{Q})$ and $\vec{Q}$ is 90°

CHEMISTRY

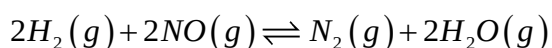
Solution 26:



The partial oxidation of glucose by Tollen's reagent or Fehling solution gives gluconic acid. It can't form hemiacetal or acetal.

Solution 27: In two beakers (I) having a pure volatile solvent and (II) having a volatile solvent and non-volatile solute. Due to non-volatile solute, vapour pressure will be lower in second beaker.

Solution 28:



$$K_c = \frac{k_f}{k_b} = \frac{[N_2][H_2O]^2}{[H_2]^2[NO]^2}$$

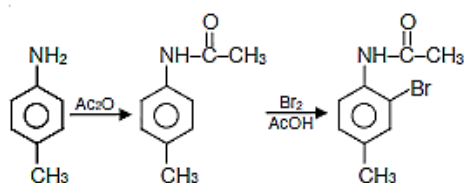
For reverse reaction rate = $k_b [N_2][H_2O]^2 / [H_2]$

Solution 29: Vinyl halides and aryl halides do not give Friedel craft's reaction because of unstable carbocation formation.

Solution 30: Bond order of CN^- is 3. It is a 14 electron system and diamagnetic in nature.

Solution 31: In sp^3 , the central metal is not chiral so does not show optical isomers. Similarly, in case of dsp^2 , two geometrical isomers will form but no optical isomers.

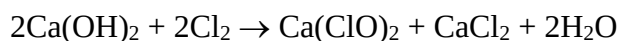
Solution 32:



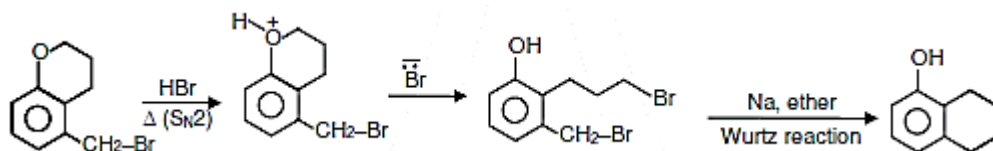
Solution 33: When Chlorine reacts with hot and conc. Sodium hydroxide to form sodium chloride and sodium chlorate.



Calcium hydroxide react with chlorine to produce bleaching powder, calcium chloride and water.



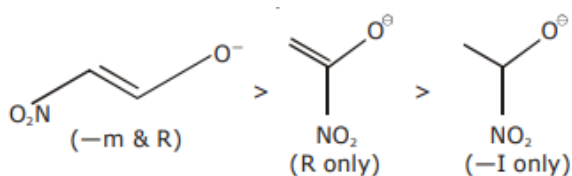
Solution 34:



Solution 35: Conceptual

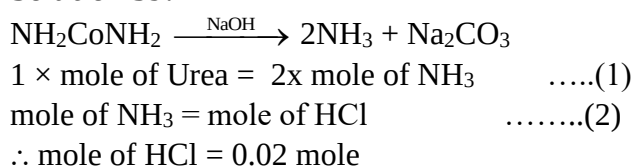
Solution 36: It depends on the Z. When Z behave as nucleophile, it will go for substitution but in case of Bulky Z behave as base and do elimination.

Solution 37:



Solution 38: Conceptual

Solution 39:



Solution 40: First more polar compound will come.

Solution 41: Conceptual

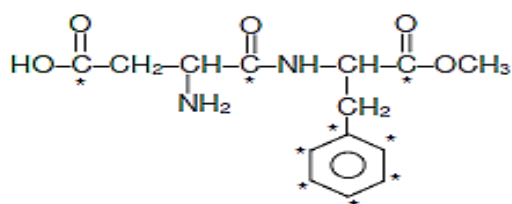
Solution 42: Conceptual

Solution 43: Conceptual

Solution 44: $C_{\text{RMS}} > C_{\text{Avg}} > C_{\text{MPS}}$

Solution 45: Conceptual

Solution 46:

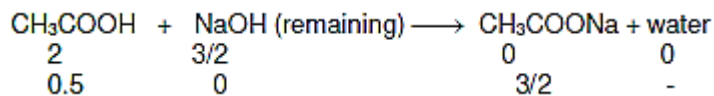


Solution 47:

m mole of acetic acid in 20 mL = 2

m mole of HCl in 20 mL = 1

m mole of NaOH = 2.5



$$\text{pH} = \text{pK}_a + \log \frac{1.5}{0.5}$$

$$= 4.75 + \log 3$$

$$= 4.75 + 0.4771 = 5.23$$

Solution 48: $2\text{C}_s + 3\text{H}_2 \rightarrow \text{C}_2\text{H}_6$

$$\Delta H_f^\circ \text{C}_2\text{H}_6 = \Delta H_c^\circ \text{Reactant} - \Delta H_c^\circ \text{product}$$

$$= 2\Delta H_c^\circ \text{C}_s + 3\Delta H_c^\circ \text{H}_2 - \Delta H_c^\circ \text{C}_2\text{H}_6$$

$$= 2 \times -286 + 3 \times (-393.5) + 1560$$

$$= -572 - 1180.5 + 1560$$

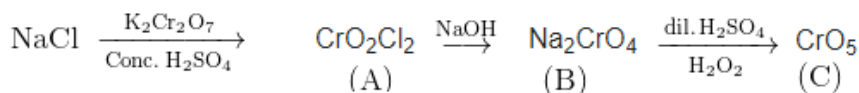
$$= -1752.5 + 1560 = -192.5 \text{ kJ}$$

Solution 49: Millimole of HCl required for 1 litre = 30

Millimole H_2SO_4 required for 1 litre = 15

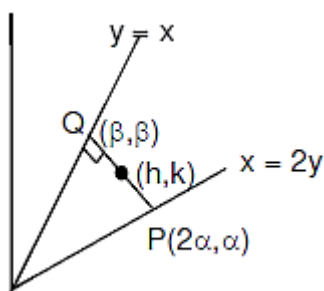
$$\text{For 250 mL of sol} = 15 \times 10^{-3} \times \frac{98}{4} = 0.37 \text{ Gram}$$

Solution 50:



MATHEMATICS

Solution 51:



$$m_{PQ} = \frac{k - \alpha}{h - 2\alpha} = -1 \Rightarrow k - \alpha = -h + 2\alpha \Rightarrow \alpha = \frac{h + k}{3} \quad \dots(1)$$

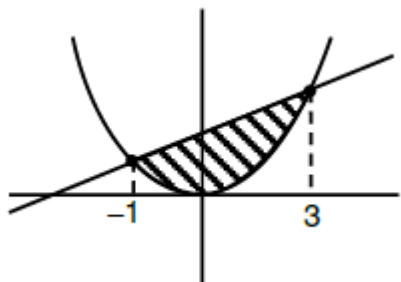
$$\text{Also, } 2h = 2\alpha + \beta \text{ and } 2k = \alpha + \beta$$

$$\therefore 2h = \alpha + 2k \Rightarrow \alpha = 2h - 2k \quad \dots(2)$$

From equations (1) and (2), we have $\frac{h+k}{3} = 2(h-k)$

So, the required locus is $6x - 6y = x + y$, that is $5x = 7y$

Solution 52:



Solving $4x^2 = y$ and $y = 8x + 12$, we get

$$4x^2 = 8x + 12 \Rightarrow x^2 - x - 3 = 0 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x+1)(x-3) = 0 \Rightarrow x = -1, 3$$

So, the required area is calculated as follows.

$$A = \int_{-1}^3 (8x + 12 - 4x^2) dx = \left[\frac{8x^2}{2} + 12x - \frac{4x^3}{3} \right]_{-1}^3 = (4(9) + 36 - 36) - \left(4 - 12 + \frac{4}{3} \right) = \frac{128}{3}$$

Solution 53:

$$z = \frac{(3 + i \sin \theta)(4 + i \cos \theta)}{(4 - i \cos \theta)(4 + i \cos \theta)}$$

Since it is purely real, imaginary part will be 0

$$\Rightarrow 3 \cos \theta + 4 \sin \theta = 0$$

$$\Rightarrow \tan \theta = -\frac{3}{4}$$

$$\text{Now, } \arg(\sin \theta + i \cos \theta) = \pi + \tan^{-1} \left(\frac{\cos \theta}{\sin \theta} \right) = \pi + \tan^{-1} \left(-\frac{4}{3} \right) = \pi - \tan^{-1} \left(\frac{4}{3} \right)$$

Solution 54:

The favourable cases are when no machine has any fault, when only one machine has a fault and when exactly two machines have faults.

So, the probability is as follows.

$$\begin{aligned} & {}^5C_0 \left(\frac{3}{4} \right)^5 + {}^5C_1 \left(\frac{1}{4} \right) \left(\frac{3}{4} \right)^4 + {}^5C_2 \left(\frac{1}{2} \right)^2 \left(\frac{3}{4} \right)^3 \\ &= \frac{243}{1024} + \frac{405}{1024} + \frac{270}{1024} = \frac{918}{1024} = \frac{459}{512} = \frac{27 \times 17}{64 \times 8} \end{aligned}$$

$$\therefore \left(\frac{3}{4}\right)^3 \times k = \left(\frac{3}{4}\right)^3 \times \frac{17}{8} \Rightarrow k = \frac{17}{8}$$

Solution 55:

Since f is a polynomial function, it is continuous and differentiable in the interval $[0, 1]$

Here, $f(0) = 11$ and $f(1) = 1 - 4 + 8 + 11 = 16$

Now, $f'(x) = 3x^2 - 8x + 8$

$$\therefore f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{16 - 11}{1} = 3c^2 - 8c + 8$$

$$3c^2 - 8c + 3 = 0 \Rightarrow c = \frac{8 \pm 2\sqrt{7}}{6} = \frac{4 \pm \sqrt{7}}{3} \Rightarrow c = \frac{4 - \sqrt{7}}{3} \in (0, 1)$$

Solution 56:

We have $\alpha^5 = 5\alpha + 3$ and $\beta^5 = 5\beta + 3$

$$P_5 = 5(\alpha + \beta) + 6 = 5(1) + 6 = 11$$

$$\text{Also, } P_5 = \alpha^2 + \beta^2 = \alpha + 1 + \beta + 1$$

$$\text{Now, } P_2 = 3 \text{ and } P_3 = \alpha^3 + \beta^3 = 2\alpha + 1 + 2\beta + 1 = 2(1) + 2 = 4$$

$$\text{So, } P_2 \times P_3 = 12$$

$$\text{Clearly, } P_5 \neq P_2 \times P_3$$

Solution 57:

$$L = \sqrt{S_1} = \sqrt{16} = 4 \text{ and } R = \sqrt{16 + 4 - 16} = 2$$

$$\text{So, length of CoC} = \frac{2LR}{\sqrt{L^2 + R^2}} = \frac{2 \times 4 \times 2}{\sqrt{16 + 4}} = \frac{16}{\sqrt{20}}, \text{ square of which is } \frac{64}{5}$$

Solution 58:

$$\text{Consider } 4\alpha \left\{ \int_{-1}^0 e^{\alpha x} dx + \int_0^2 e^{-\alpha x} dx \right\} = 5$$

$$\Rightarrow 4\alpha \left\{ \left(\frac{e^{\alpha x}}{\alpha} \right)_{-1}^0 + \left(\frac{e^{-\alpha x}}{-\alpha} \right)_0^2 \right\} = 5 \Rightarrow 4\alpha \left\{ \left(\frac{1 - e^{-\alpha}}{\alpha} \right) - \left(\frac{e^{-2\alpha} - 1}{\alpha} \right) \right\} = 5 \Rightarrow 4(2 - e^{-\alpha} - e^{-2\alpha}) = 5$$

Now, substituting $e^{-\alpha} = t$, we get the following.

$$\Rightarrow 4t^2 = 4t - 3 = 0$$

$$\Rightarrow (2t + 3)(2t - 1) = 0$$

$$\Rightarrow e^{-\alpha} = \frac{1}{2}$$

$$\Rightarrow \alpha = \ln 2$$

Solution 59:

Given $x = \frac{1}{2}$ and $y = \frac{-1}{4}$, so $xy = \frac{-1}{8}$

Now, consider $y \times \frac{1(-2x)}{2\sqrt{-x^2}} + y' \times \sqrt{1-x^2} = - \left\{ 1 \times \sqrt{1-y^2} + \frac{x \times (-2y)}{2\sqrt{1-y^2}} y' \right\}$

That is, $-\frac{xy}{\sqrt{1-x^2}} + y' \sqrt{1-x^2} = -\sqrt{1-y^2} + \frac{xyy'}{\sqrt{1-y^2}}$

or $y' \left(\sqrt{1-x^2} - \frac{xy}{\sqrt{1-y^2}} \right) = \frac{xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$

So, $y' \left(\frac{\sqrt{3}}{2} + \frac{1}{8 \times \frac{\sqrt{15}}{4}} \right) = \frac{-1}{8 \times \frac{\sqrt{3}}{2}} - \frac{\sqrt{15}}{4}$

This gives us $y' \left(\frac{\sqrt{45}+1}{2\sqrt{15}} \right) = -\frac{(1+\sqrt{45})}{4\sqrt{3}} \Rightarrow y' = -\frac{\sqrt{5}}{2}$

Solution 60:

Consider the equation

$$\frac{36}{r+1} \times \frac{35}{C_r} (k^2 - 3) = \frac{35}{C_r} \Rightarrow k^2 - 3 = \frac{r+1}{6} \Rightarrow k^2 = 3 + \frac{r+1}{6}$$

Here, r can be 5, 35

$$r = 5 \Rightarrow k = \pm 2 \text{ and } r = 35 \Rightarrow k = \pm 3$$

Hence, the number of order pairs is 4

Solution 61:

From the data in the question, we have

$$a_1 + a_2 = 4 \Rightarrow a_1 + a_1 r = 4 \quad \dots (i)$$

$$a_3 + a_4 = 16 \Rightarrow a_1 r^2 + a_1 r^3 = 16 \quad \dots (ii)$$

Solving these equations, we get $\frac{1}{r^2} + \frac{1}{4} \Rightarrow r^2 = 4 \Rightarrow r = \pm 2$

$$r = 2 \Rightarrow a_1(1+2) = 4 \Rightarrow a_1 = \frac{4}{3}$$

$$r = -2 \Rightarrow a_1(1-2) = 4 \Rightarrow a_1 = -4$$

$$\text{Now, } \sum_{i=1}^a a_i = \frac{a_1(r^q - 1)}{r - 1} = \frac{(-4)((-2)^9 - 1)}{-2 - 1} = \frac{4}{3}(-513) = 4\lambda$$

$$\text{So, } \lambda = -171$$

Solution 62:

Let $P = A \subset B$, $Q = B \subset C$ and $R = C \subset A$

Now, the contrapositive of $(P \wedge Q) \rightarrow R$ is $\sim R \rightarrow \sim(P \wedge Q)$,

That is, $\sim R \rightarrow \sim(P) \vee \sim(Q)$

Solution 63:

$$\text{Consider } 2\cot^2 \theta - \frac{5}{\sin \theta} + 4 = 0 \Rightarrow \frac{2\cos^2 \theta}{\sin^2 \theta} - \frac{5}{\sin \theta} + 4 = 0$$

$$\text{That is, } 2\cos^2 \theta - 5\sin \theta + 4\sin^2 \theta = 0, \sin \theta \neq 0$$

$$\Rightarrow 2\sin^2 \theta - 5\sin \theta + 2 = 0 \Rightarrow (2\sin \theta - 1)(\sin \theta - 2) = 0$$

$$\therefore \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} \therefore \int_{\pi/6}^{5\pi/6} \cos^2 3\theta d\theta &= \int_{\pi/6}^{5\pi/6} \frac{1 + \cos 6\theta}{2} d\theta \\ &= \frac{1}{2} \left[\theta + \frac{\sin 6\theta}{6} \right]_{\pi/6}^{5\pi/6} \\ &= \frac{1}{2} \left[\frac{5\pi}{6} - \frac{\pi}{6} + \frac{1}{6}(0 - 0) \right] \\ &= \frac{1}{2} \times \frac{4\pi}{6} \\ &= \frac{\pi}{3} \end{aligned}$$

Solution 64:

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 0 \Rightarrow 3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\text{So, } (\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = \frac{-3}{2} \Rightarrow \lambda = \frac{-3}{2}$$

$$\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times (-\vec{a} - \vec{b}) + (-\vec{a} - \vec{b}) \times \vec{a} = \vec{a} \times \vec{b} + \vec{a} \times \vec{b} + \vec{a} \times \vec{b}$$

$$\text{So, } \vec{d} = 3(\vec{a} \times \vec{b})$$

Solution 65:

$$\text{Consider } f(x) = ax^5 + bx^4 + cx^3$$

$$\lim_{x \rightarrow 0} \left(2 + \frac{ax^5 + bx^4 + cx^3}{x^3} \right) = 4 \Rightarrow 2 + c = 4 \Rightarrow c = 2$$

$$\text{Now, } f'(1) = 0 \Rightarrow 5a + 4b + 6 = 0 \text{ and } f'(-1) = 0 \Rightarrow 5a - 4b + 6 = 0$$

$$\text{Solving these, we get } a = -\frac{6}{5} \text{ and } b = 0$$

$$\text{So, } f(x) = \frac{-6}{5}x^5 + 2x^3$$

$$\therefore f'(x) = -6x^4 + 6x^2 = 6x^2(-x^2 + 1) = -6x^2(x+1)(x-1)$$

$$\begin{array}{ccc} -1 & + & 1- \\ \hline 1- & & 1 \end{array}$$

Clearly, f has a minimum at $x = -1$ and a maximum at $x = 1$

Solution 66:

$$3x + 4y = 12\sqrt{2} \Rightarrow 4y = -3x + 12\sqrt{2} \Rightarrow y = -\frac{3}{4}x + 3\sqrt{2}$$

For tangency, we have $c^2 = a^2m^2 + b^2$. So, we get the following.

$$18 = a^2 \times \frac{9}{16} + 9 \Rightarrow a^2 \times \frac{9}{16} = 9 \Rightarrow a^2 = 16 \Rightarrow a = 4$$

$$\text{Now, eccentricity } e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$$

$$\therefore ae = \frac{\sqrt{7}}{4} \times 4 = \sqrt{7}$$

So, the foci are $(\pm\sqrt{7}, 0)$ and hence, the distance between foci = $2\sqrt{7}$

Solution 67:

The equation $\frac{dy}{dx} + x = y^2$ is a linear differential equation.

The integrating factor is $IF = e^{\int 1 \cdot dy} = e^y$.

So, the solution is given by $.e^y = \int y^2 e^y dy = y^2 e^y - \int 2ye^y dy$

$$\Rightarrow xe^y = y^2 e^y - 2(ye^y - e^y) + C \Rightarrow xe^y = y^2 e^y - 2ye^y + 2e^y + C \Rightarrow x = y^2 = 2y + 2 + ce^{-y}$$

To find c , we substitute $y = 1$ at $x = 0$

$$\text{This gives us } 0 = 1 - 2 + 2 + \frac{c}{e} \Rightarrow c = -e$$

So for $y = 0$, we have

$$x = 0 - 0 + 2 = (-e)(e^{-0}) \Rightarrow x = 2 - e$$

Solution 68:

$$|B| = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} = \begin{vmatrix} 3^0 a_{11} & 3^1 a_{12} & 3^2 a_{13} \\ 3^1 a_{21} & 3^2 a_{22} & 3^3 a_{23} \\ 3^2 a_{31} & 3^3 a_{32} & 3^4 a_{33} \end{vmatrix} \Rightarrow 81 = 3^3 \cdot 3 \cdot 3^2 |A| \Rightarrow 3^4 = 3^6 |A| \Rightarrow |A| = \frac{1}{9}$$

Solution 69:

Given series is $S = 3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \dots$ up to 40 terms

That is, $S = 7 + 17 + 27 + 37 + 47 + \dots$ up to 20 terms

$$S_{40} = \frac{20}{2} [2 \times 7 + (19)10] = 10[14 + 190] = 10[2040] = (102)(20)$$

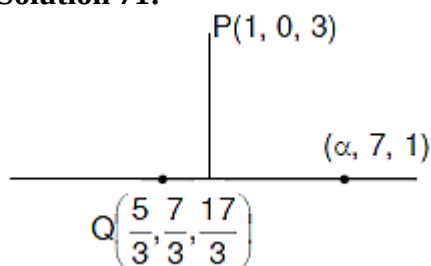
$$\therefore m = 20$$

Solution 70:

$$\text{Consider } \frac{(1+x)^{10} \left[1 - \left(\frac{x}{1+x} \right)^{11} \right]}{\left(1 - \frac{x}{1+x} \right)} = \frac{(1+x)^{10} [(1+x)^{11} - x^{11}]}{(1+x)^{11} \times \frac{1}{(1+x)}} = (1+x)^{11} - x^{11}$$

$$\text{Coefficient of } x^7 \text{ is } {}^{11}C_7 = {}^{11}C_4 = 330$$

Solution 71:



The line segment PQ is perpendicular to the line L.

So, we have the following condition.

$$\begin{aligned} \left(1 - \frac{5}{3}\right)\left(\alpha - \frac{5}{3}\right) + \left(\frac{-7}{3}\right)\left(7 - \frac{7}{3}\right) + \left(3 - \frac{17}{3}\right)\left(1 - \frac{17}{3}\right) &= 0 \\ \Rightarrow \frac{-2\alpha}{3} + \frac{10}{9} - \frac{98}{9} + \frac{112}{9} &= 0 \\ \Rightarrow \frac{2\alpha}{3} &= \frac{24}{9} \\ \Rightarrow \alpha &= 4 \end{aligned}$$

Solution 72:

$$\begin{aligned} \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \left(\frac{1}{x} \ln \left(\frac{1+3x}{1-2x} \right) \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{\ln(1+3x)}{x} - \frac{\ln(1-2x)}{x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{3\ln(1+3x)}{3x} - \frac{2\ln(1-2x)}{-2x} \right) \\ &= 3 + 2 = 5 \end{aligned}$$

So, for $f(x)$ to be continuous if $f(0) = \lim_{x \rightarrow 0} f(x) = 5$

Solution 73:

$$\bar{x} = \frac{3+7+9+12+13+20+x+y}{8} = 10 \Rightarrow x+y=16 \quad \dots(i)$$

$$\begin{aligned} \sigma^2 &= \frac{\sum (x_i)^2}{8} - (\bar{x})^2 = 25 \\ \Rightarrow \frac{9+49+81+144+169+400+x^2+y^2}{8} - 100 &= 25 \Rightarrow x^2+y^2=148 \quad \dots(ii) \end{aligned}$$

Now, consider the following

$$(x+y)^2 = (16)^2 \Rightarrow x^2 + y^2 + 2xy = 256 \Rightarrow xy = 54$$

Solution 74:

The given system of equations may be interpreted as 3 planes in 3D space as follows.

$$x + y + z = 6 \quad \dots(1)$$

$$x + 2y + 3z = 10 \quad \dots(2)$$

$$3x + 2y + \lambda z = \mu \quad \dots(3)$$

For planes (1) and (2), considering $y=0$ gives us $x+z=6$ and $x+3z=10$

Solving, we get $x = 4$ and $z = 2$, which represents the point $(4, 0, 2)$.

For planes (1) and (2), considering $z = 0$ gives us $x + y = 6$ and $x + 2y = 10$

Solving, we get $x = 2$ and $y = 4$, which represents the point $(2, 4, 0)$.

So, the plane $3x + 2y + \lambda z = \mu$ must pass through both $(2, 4, 0)$ and $(4, 0, 2)$

This gives us $6 + 8 = \mu \Rightarrow \mu = 14$

and $12 + 2\lambda = \mu$, that is, $12 + 2\lambda = 14 \Rightarrow \lambda = 1$

Thus, $\mu - \lambda^2 = 14 - 1 = 13$

Solution 75:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 25 + 7 - 3 = 29$$